SOUMYAJIT SARKAR

Full Stack Developer | Machine Learning Engineer

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SUMMARY

Software Development Engineer with 2 years of industry experience, skilled in Java, Python, and the MERN stack, with a strong foundation in building scalable, production-ready applications. Currently pursuing a Master's in Computer Applications (AI/ML specialization), with hands-on experience in machine learning, deep learning, and data-driven problem solving. Experienced in model development, data preprocessing, and integrating intelligent solutions into real-world systems. Adept at writing clean, maintainable code and collaborating within cross-functional teams. Passionate about bridging software engineering and AI to deliver impactful, scalable solutions.

PROFESSIONAL EXPERIENCE

Murphi.ai (Deskfactors) — NOV 2025 - JAN 2026

- **AI/ML Intern**
- **Project → ICD-10**
  - Worked on a Medical Coding (ICD-10) AI project, contributing to end-to-end system architecture design for automated clinical coding workflows.
  - Implemented and optimized keyphrase extraction pipelines from clinical narratives to support accurate ICD-10 code prediction.
  - Refined and engineered prompts for LLM-based medical coding assistants to improve coding relevance, consistency, and compliance.
  - Performed L1 medical coding validation, testing AI-generated ICD-10 codes against clinical documentation to ensure accuracy and guideline adherence.
- **Project → Sola Payments**
  - Designed the end-to-end UI/UX for Sola Payments, an AI-powered payment chatbot, focusing on user flow, conversational design, and accessibility.
  - Gained hands-on experience in AI/ML product development, healthcare NLP, prompt engineering, and UX design integration.

Cognizant Technology Solutions — AUG 2022 - MAY 2024

- **Software Developer**
- Offering Portfolio → Pega Developer
- **Project → AIG Japan**
  - Generated reports that improved decision-making policy and claims management.
  - Identified and resolved critical issues using Tracer tool, Clipboard, reducing system downtime.
  - Enhanced business process by optimizing Pega workflows and automations.
  - Attended client meetings and implemented real-time solutions based on evolving requirements.

ACADEMIC & PERSONAL PROJECTS

- **Deep Learning Projects**
  - **Chest Cancer Detection**
    - Developed an image classification model which classifies the type of chest cancer from x-ray images
    - **Techstack** → Python, Pandas, matplotlib, EfficientNetB0, Scikit-Learn, Grad-Cam, OpenCV.
    - **Impact** → Helps in faster and more accurate chest cancer diagnosis from X-ray images.
  - **Object Detection System**
    - Developed an AI-powered model that detects and classifies objects in images with high accuracy.
    - **Techstack** → Python, matplotlib, torchvision, Open Images Dataset.
    - **Impact ->** Enables faster and smarter decision-making by automatically identifying and classifying objects from images.
  - **Smart Healthcare System**
    - A recommendation system that provides medicines, precautions, diet plans and disease insights based on the given user input.
    - **Techstack** → Python, Pandas, SVC, Random Forest Regressor, Logistic Regression
    - **Impact** → Helps in early prevention through recommendations before reaching out to a hospital.
  - **Speech Emotion Recognition**
    - A recommendation system that analyzes voice recordings to detect human emotions.
    - Techstack → Flask, MySQL, Python, Pandas, TensorFlow, Keras, Librosa
    - Impact → Enables emotion-aware systems by analyzing user audio input to accurately classify tones such as anger, happiness, love, and grief for enhanced healthcare and user experience applications.
  - **AI Trip Planner**
    - AI-powered travel assistant that streamlines trip planning with destination insights, real-time weather updates, budget estimation, and integrated currency conversion.
    - **Techstack** → JavaScript, Python, REST APIs
    - **Framework** → React.js, Tailwind CSS, FastAPI (served via Uvicorn)
    - **Impact** → Makes travel planning faster and more organized.
- **Full Stack Projects**
  - **Ping-Ev**
    - Ping-EV is a smart EV charging assistant that proactively alerts drivers of low battery levels and seamlessly guides them to the nearest charging stations, ensuring a reliable and stress-free driving experience.
    - **Techstack** → Typescript, HTML, CSS, Javascript
    - **Framework** → React
    - **Impact** → Reduces range anxiety by enabling timely charging decisions and ensuring uninterrupted, confident EV journeys.

- **LRU Cache Visualizer**
  - Implemented an LRU Cache using a HashMap and doubly linked list to achieve O(1) access and efficient eviction of least recently used items within a fixed capacity.
  - **Techstack** → Typescript, HTML, CSS, Javascript
  - **Framework** → React
  - **Impact** → Helps the learners to visualize how LRU cache works.

CERTIFCATIONS

- C Certified Pega System Architect (CSA) 8.7 version
- Certified Pega Senior System Architect (CSSA) 8.8 version
- Fundamentals of Accelerated Computing with CUDA C/C++ (NVIDIA)
- Fundamentals of Accelerated Computing with CUDA Python (NVIDIA)

EDUCATION

JAIN UNIVERSITY (Bengaluru) **2024 - 2026**

- Master of Computer Application (**Specialization in AI/ML**)
  - GPA → 8.28

INSTITUTE OF ENGINEERING & MANAGEMENT (Kolkata) **2019 - 2022**

- Bachelor of Computer Application
  - GPA → 9.59

SKILLS & TOOLS

- **Programming Languages:** Java, Python, C++, C, JavaScript, TypeScript
- **Frontend:** HTML, CSS, JSP, JavaScript
- **Programming, Data Handling & Visualization:** Python, NumPy, Pandas, Matplotlib, Seaborn, Power BI
- **Deep Learning:** TensorFlow, Keras (Sequential, ResNet50, MobileNetV2, EfficientNetB0)
- **Frameworks & Libraries:** Spring, SpringBoot, React, Hibernate, FastAPI, Streamlit, Flask, TailWind CSS
- **Database:** MySQL, Oracle SQL, MongoDB
- **Tools & Platform:** Git, Postman, Docker
- **Core Technical Skills:** Data Structures & Algorithms, Problem Solving, Core JAVA, Object Oriented Programming(OOP)
- **Management Skills:** Project Management
- **AI Tools:** Claude Code, GitHub Copilot, Gemini, Antigravity
- **Languages:** English (Fluent), Hindi (Fluent), Bengali (Fluent)

ACHIEVEMENTS

- Pitched AI-based startup **Athlink** with team **Startup Visionaries** at a business ideation event (Feb 2025), ranked **Top 5**, focused on empowering underprivileged athletes.